Assignment - 1

Finding the Rank of a Matrix

1. Find the rank of the following matrices using elementary row operations.

1. $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ (2×3)

2. $B = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 4 \\ 2 & 1 & 0 \end{bmatrix}$ (3×3)

3. $C = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 4 & 5 & 6 & 7 \\ 1 & 0 & 1 & 2 \end{bmatrix}$ (3×4)

4. $D = \begin{bmatrix} 2 & 1 & -1 & 0 \\ 0 & 3 & 1 & 2 \\ 1 & 0 & 2 & -1 \\ 4 & 1 & 0 & 3 \end{bmatrix}$ (4×4)

5. $E = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 6 & 8 & 10 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 3 & 6 & 9 & 12 & 15 \end{bmatrix}$ (5×5)

6. $F = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 \end{bmatrix}$ (3×4)

7. $G = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \\ 3 & 0 & 1 & 2 \end{bmatrix}$ (4×4)

8. $H = \begin{bmatrix} 1 & 2 & 0 & 3 & 1 \\ 0 & 1 & 4 & 2 & 0 \\ 2 & 0 & 1 & 0 & 1 \\ 1 & 3 & 2 & 1 & 0 \\ 0 & 1 & 0 & 2 & 1 \end{bmatrix}$ (5×5)

2. Find the Rank with the Help of Minor.

9. $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 2 & 1 & 1 \end{bmatrix}$ (3×3)

10. $B = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 3 & 2 \\ 0 & 1 & 1 \end{bmatrix}$ (3×3)

3. Reduce the following matrices to normal form (3×3).

11. $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 0 & 1 \end{bmatrix}$ (3×3)

12. $B = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 4 \\ 2 & 1 & 1 \end{bmatrix}$ (3×3)

13. $C = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 2 & 6 \\ 6 & 3 & 9 \end{bmatrix}$ (3×3)

14. $D = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}$ (3×3)